

GOPI JAGADHEESH MAHAMKALI

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EDUCATION

Rajiv Gandhi University of Knowledge Technologies, Nuzvid

Bachelor of Technology in Computer Science & Engineering (Expected May 2027)

September 2023 – Present

CGPA: 8.92

Rajiv Gandhi University of Knowledge Technologies, Nuzvid

Pre-University Certificate (PUC)

December 2021 – August 2023

CGPA: 9.85

WORK EXPERIENCE

4SightAI

October 2024 – March 2025

AI Engineer Intern

Vijayawada, Full-Time

- Led end-to-end development of **DOCS2DATA**, an AI-powered use case in **AI4Andhra Police Pilot Project**, for extracting structured data from police department documents, reducing manual processing by **90%**.
- Designed and implemented **AI-driven document understanding pipelines** combining Azure Document Intelligence, rule-based validation to extract, clean, and normalize unstructured data across diverse formats.
- Built and deployed scalable **FastAPI-based microservices** enabling document ingestion, processing, and retrieval, integrated within a distributed system architecture.
- Developed and integrated **signature and stamp detection models** using custom trained model based on yolo architecture to validate authenticity of official documents.
- Built a **multi-agent AI system** for road safety analysis using **YOLO + Google Gemini**, enabling contextual understanding of roadway incidents for project **RTGS** as a proof of concept.
- Designed **6 specialized AI agents** for detecting fire, water hazards, potholes, heavy vehicles, waste, and fallen trees, combining computer vision with LLM-based reasoning.

PROJECTS

Drone-Based Weapon Detection System — Team of 2 — Python, PyTorch, YOLOv8, FastAPI, Google Colab

- Designed a **two-stage detection pipeline** using **YOLOv8n** for person detection followed by a custom-trained model to identify weapons within detected human regions.
- Improved detection relevance by restricting inference to **person bounding boxes**, reducing false positives from background objects in surveillance footage.
- Developed **FastAPI backend APIs** for image/video inference, returning structured detection outputs (class labels, confidence scores, and bounding box coordinates) for downstream integration.

KAMAI – AI Financial Companion for Gig Workers — Python, FastAPI, Azure OpenAI, AutoGen, Supabase

- Built a **multi-agent AI financial companion** targeting India's **15M+ gig workers**, orchestrating **12 specialized AutoGen agents** powered by **Azure OpenAI GPT-4** for income pattern analysis, volatility forecasting, and risk assessment.
- Designed an **adaptive budgeting engine** with Feast/Famine/Normal income modes, automated bill prioritization, and integrated matching across **200+ government schemes** for personalized savings and investment planning.
- Developed full-stack application with **React 18 + TypeScript** frontend and **FastAPI** backend, featuring real-time transaction categorization, ITR filing assistance, and row-level security via **Supabase (PostgreSQL)**.

TECHNICAL SKILLS

Languages: Python, C++, SQL.

AI : Machine Learning, Deep Learning, Computer Vision, Generative AI, RAG Pipelines, Agentic AI, MCP.

Frameworks & Libraries: PyTorch, LangChain, LangGraph, FastAPI, HuggingFace, scikit-learn, OpenCV.

DevOps: Docker, Git, GitHub, Azure DevOps.

Tools: Google Colab, Kaggle, VPS, Azure Foundry, RoboFlow, Jupyter Notebook, VS Code, CursorAI, Claude.

ACHIEVEMENTS

- Secured **All India Rank 3863** in **GATE Data Science & AI (2026)**.
- Secured **1st place** in **AI 4 Andhra Police Hackathon** for **DOCS2DATA** — GenAI-based document workflow automation.
- Finalist in **Hackrx 6.0** — GenAI hackathon conducted by Bajaj Finserv.
- Won **Best UI/UX Design and Implementation Award** at **Hack To Crack 2.0** (computer-vision / analytics demo).
- Ranked **1st** in **SIH Internal Hackathon**.

CERTIFICATIONS

- Complete Data Science, Machine Learning, Deep Learning, and NLP Bootcamp** — KrishNaik Academy
- Complete Computer Vision Bootcamp** — KrishNaik Academy
- Complete Generative AI Course With Langchain and Huggingface** — KrishNaik Academy